CHAPTER 4

STD X

MATHEMATICS OF CHANCE

1) There are 7 blue balls and 8 green balls in a bag 6 blue balls and 14 green balls in another.
a) What is the probability of getting a blue ball from the first bag?
b) From the second bag?
c) If all the balls are put in a single bag, what is the probability of getting a green ball from it?

2) There are 4 apples and 6 mangoes in a bag. In another bag, there are 8 apples and 7mangoes. Which bag is better for getting an apple?

3) The counting numbers from 1 to 10 are written on different paper slips and put in a box. In another box prime numbers from 1 to 20 are written on different paper slips. If one paper slip from each box is taken at a time
1) What is the probability of both are odd numbers
2) What is the probability of the sum of the numbers being even?
3) What is the probability of the product being odd?

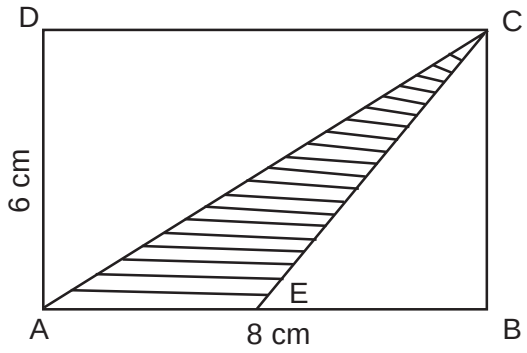
4) A box contains 50 beads of different colours white, black and red. The probability of getting a white bead is 1/5. The probability of getting a red bead is 8/25.
a) How many white beads are there?
b) What is the probability of getting a black bead from it?
c) How many more red balls should be added so that the probability of getting a red ball is 1/2?

5) 4 cards with numbers 1, 3, 5, 7 written on them are arranged in various ways without repetation.
a) What is the probability that the number is greater than 7000?
b) What is the probability that the number is less than 7000?

6) A box contains 50 bulbs, out of which 20 are fused. Another box contains 40 bulbs out of which 15 are fused. If one bulb from each box is taken at a time?
a) What is the total number of pairs?
b) What is the probability that both are working bulbs?
c) What is the probability of getting one working bulb and one fused bulb?
d) What is the probability of getting at least one fused bulb?

7) There are 8 red balls and some blue balls in a box. When a ball is drawn from this the probability of getting a red ball is 3 times the probability of getting a blue ball. How many blue balls are there?

8)



In the figure, E is the midpoint of AB. If we put a dot in the figure without looking into it.

a) What is the probability that the dot to be within the shaded region?

b) What is the probability that the dot to be within the shaded region?

ANSWER KEY

1. a) 7/15
   b) 3/10
   c) 22/35

2. Second bag

3. a) 7/16
   b) 1/2
   c) 7/16

4. a) 10
   b) 12/25
   c) 18

5. a) 1/4
   b) 3/4

6. a) 3000
   b) 9/20
   c) 9/20
   d) 11/20

7. 2 എണ്ണം

8. a) 1/4
   b) 3/4

PREPARED BY :

SEEMA SUGATHAN
HST (Maths)
GHS Kanichukulangara